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## Topic 18

# Monitoring & Control

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Monitoring and control are essential aspects of managing a software project. These processes help ensure that the project stays on track, meets its objectives, and delivers quality outcomes within time and budget constraints.

### 1. Performance Measurement

Performance measurement is the process of assessing the progress of the project based on predefined metrics and key performance indicators (KPIs). It involves tracking and comparing actual performance with the project plan, identifying variances, and taking corrective actions when necessary.

*Key Performance Indicators (KPIs):*

KPIs help in measuring the health and success of a project. Some of the common KPIs used in software project management include:

- **Schedule Variance (SV):** Measures the difference between the planned progress and the actual progress.
  - Formula:  $SV = EV \text{ (Earned Value)} - PV \text{ (Planned Value)}$
- **Cost Variance (CV):** Measures the difference between the actual costs and the planned costs.
  - Formula:  $CV = EV - AC \text{ (Actual Cost)}$
- **Earned Value (EV):** The value of work actually performed compared to what was planned.
- **Planned Value (PV):** The value of the work that was planned to be completed by a certain point.
- **Actual Cost (AC):** The actual cost incurred for the work completed.

*Example of Performance Measurement:*

Imagine a software project where the initial schedule was to complete 100 features in 6 months. After 3 months, the team has completed only 40 features, but 50 were planned to be done by then. By using performance measurements like **Schedule Variance (SV)** and **Earned Value (EV)**, you can determine how much behind the schedule the project is.

### 2. Change Control Processes

Change control refers to the procedures for managing changes to the project's scope, schedule, or resources. It ensures that no unauthorized changes are made and that any changes are properly assessed and documented.

### *Key Steps in Change Control:*

1. **Request for Change:** Any stakeholder requests a change to the project, such as adding new features or adjusting deadlines.
  - Example: A client asks to add a new payment gateway after the initial project requirements were finalized.
2. **Impact Analysis:** The project team assesses how the proposed change will impact the project in terms of scope, schedule, cost, and quality.
  - Example: The new payment gateway requires additional development and testing, which will extend the project timeline by 2 weeks and increase costs by \$5000.
3. **Approval or Rejection:** The change request is evaluated by a change control board or project manager, who decides whether the change should be approved or rejected.
  - Example: After careful review, the change control board approves the new payment gateway feature but requests that the additional costs be covered by the client.
4. **Implementation:** Once approved, the changes are implemented, and the project plan is updated accordingly.
  - Example: Developers start working on integrating the new payment gateway, and the project timeline is adjusted to reflect the delay.
5. **Communication:** The project team communicates the changes to all stakeholders, including the updated project schedule and cost estimates.
  - Example: The project manager sends a formal email to all team members, updating them on the revised project schedule and highlighting the new feature requirements.
6. **Documenting Changes:** The changes are documented in the project management system for future reference and transparency.
  - Example: The change request, its impact analysis, and approval documents are uploaded to the project's management software for record-keeping.

### *Example of Change Control Process:*

A project for developing a mobile app was initially scoped to include basic functionality. Midway through development, the client requests a new feature to allow the app to integrate with third-party services. The project manager follows the change control process by documenting the request, assessing its impact on the budget and schedule, and presenting it to the change control board for approval.

## **Scenario-Based Questions and Solutions**

### *Scenario 1:*

**Question:** A software project is running behind schedule, and there are constant changes in the requirements. The team is frustrated, and the client is dissatisfied with the delays. How would you apply performance measurement and change control to address this issue?

### **Solution:**

1. **Performance Measurement:**
  - First, calculate **Schedule Variance (SV)** and **Cost Variance (CV)** to assess the project's current status. If there is a significant gap between actual progress and planned progress, the project team needs to identify bottlenecks and areas for improvement.

- Use **Earned Value (EV)** and **Planned Value (PV)** to understand how much work has been completed compared to what was planned.
- Take corrective actions like reassigning resources, streamlining workflows, or adjusting the project schedule.

## 2. **Change Control:**

- Any changes to the project's requirements must be properly documented and assessed for their impact on the project timeline and costs.
- Conduct a thorough **Impact Analysis** for each new requirement, determining how it will affect the overall project.
- Submit the change request for approval and communicate any changes to the stakeholders, ensuring that everyone is on the same page.

By applying both performance measurement and change control processes, you can identify the root cause of delays and ensure that any changes are managed effectively to avoid further schedule impacts.

### *Scenario 2:*

**Question:** After implementing a new feature in the software, the testing phase reveals that the feature does not work as expected. The client requests additional changes, which will require more resources and time. How would you handle this using change control processes?

### **Solution:**

1. **Request for Change:** The client formally requests the new changes after testing reveals issues with the feature.
  - The project manager should create a **Change Request** document to describe the issue and the requested changes.
2. **Impact Analysis:** The team assesses how the changes will affect the project. This includes additional resources required, the extension of the timeline, and any impact on the project budget.
  - Example: The testing team identifies that the feature requires rework and additional integration, which will delay the timeline by 3 weeks.
3. **Approval or Rejection:** The change control board or project manager decides whether the changes are feasible based on the impact analysis.
  - If approved, the team can begin working on the changes, but they must also get confirmation from the client on additional costs.
4. **Implementation:** Once approved, the development and testing teams proceed with the required changes, and the project schedule is updated to reflect the new timeline.
5. **Communication and Documentation:** The changes are communicated to all stakeholders, and the project documentation is updated to reflect the modifications.

By following the change control process, the project team ensures that the requested changes are properly evaluated, managed, and documented, avoiding chaos and keeping the project on track.

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